Skin Lesion Classification (Benign vs Malignant)

ITAI 1378 Final Project

(This project is for educational purposes only and is not a medical diagnostic tool.)

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Skin Lesion Classification (Benign vs Malignant)

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- Diane Lenhoff

- ITAI 1378 - Final Project
- Tier 1 Image Classification

The Problem

- Skin cancer can be difficult to detect early, and delayed diagnosis can lead to serious health risks.
- Doctors, dermatologists, and patients benefit from faster screening tools that can help identify suspicious lesions earlier.
- Early detection improves treatment outcomes and reduces health risks.

Proposed Solution

Build a computer vision system that:

- Takes an image of a skin lesion
- Uses a pre-trained ResNet18 model
- Classifies the lesion as benign or malignant
- Provides a confidence score for prediction

- Image → Preprocessing → Model → Prediction → Results

Technical Approach

Computer Vision Technique

Image Classification

Model Used

ResNet18 with Transfer Learning

Frameworks

- PyTorch
- Torchvision
- Google Colab
- Scikit-learn
- Matplotlib

Why This Approach

Transfer learning improves performance with smaller datasets and faster training.

Dataset Used

Source

Public Kaggle Dataset:
Skin Cancer – Malignant vs Benign

Dataset Size

- Training Images: 2,637
- Testing Images: 660

Labels

- Benign
- Malignant

Preprocessing

- Resize to 224x224
- Normalize images
- Random flip + rotation
- Train/Test split

System Architecture

System Flow

Input Image
↓
Resize + Normalize
↓
ResNet18 Model
↓
Prediction
↓
Benign or Malignant
↓
Evaluation Metrics

Evaluation includes:

- Accuracy
- Confusion Matrix
- Precision / Recall / F1-score

Model Training

Training Setup

- Transfer Learning with ResNet18
- Optimizer: Adam
- Initial Learning Rate: 0.001
- Improved Learning Rate: 0.0001
- Epochs: 10 total
- GPU used in Google Colab

Improvement

Initial test accuracy: 69.24%
Final test accuracy: 90.15%

Final Results

Final Test Accuracy

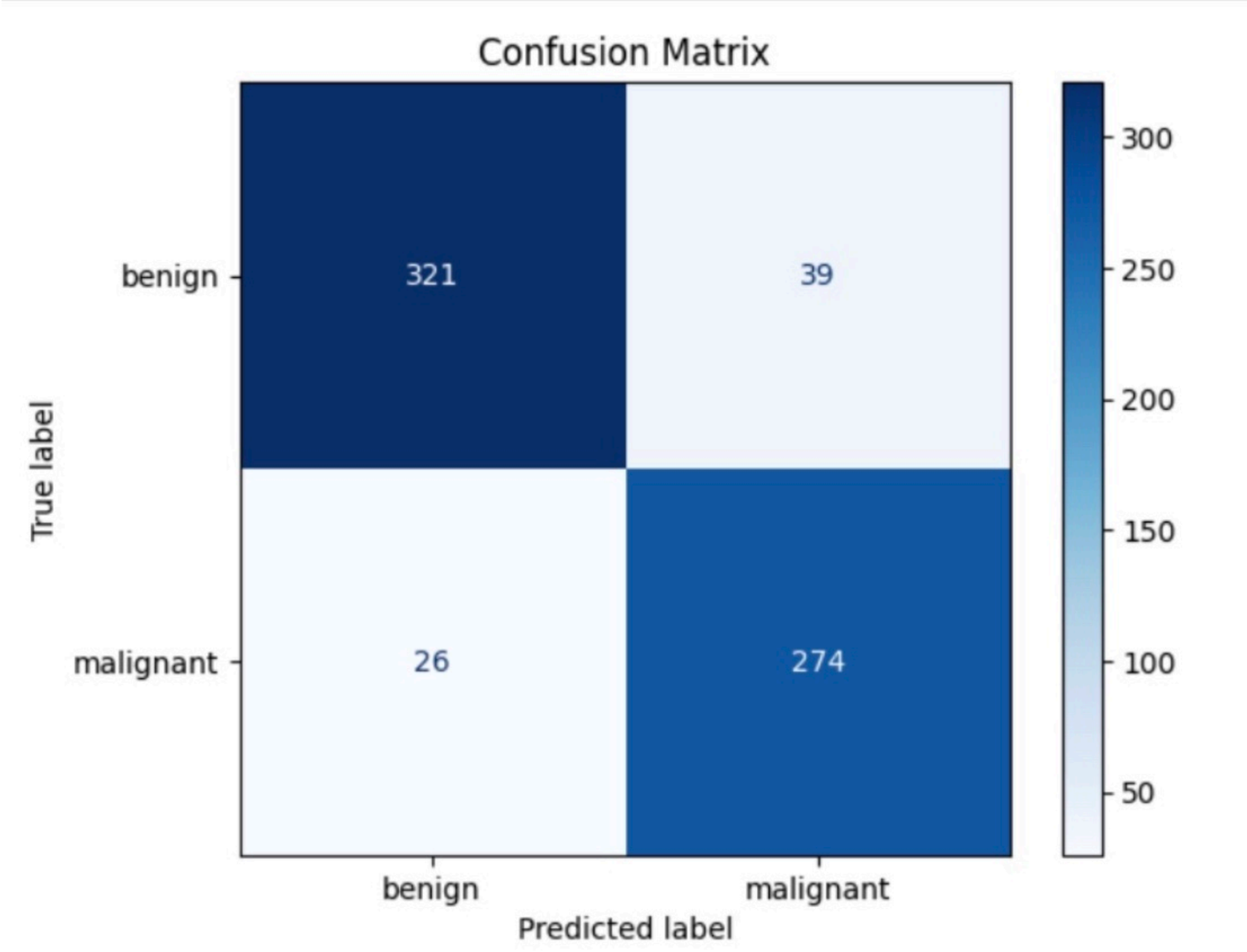
90.15%

Additional Metrics

- Precision
- Recall
- F1-score
- Confusion Matrix
- Confidence Scores

Speed Goal

Less than 1 second per image



Challenges & Solutions

Challenges and Fixes

Challenge	Solution
Low initial accuracy	Lowered learning rate and trained additional epochs
Class imbalance	Used image augmentation
Limited training time	Used pretrained model instead of training from scratch

This improved model performance significantly.

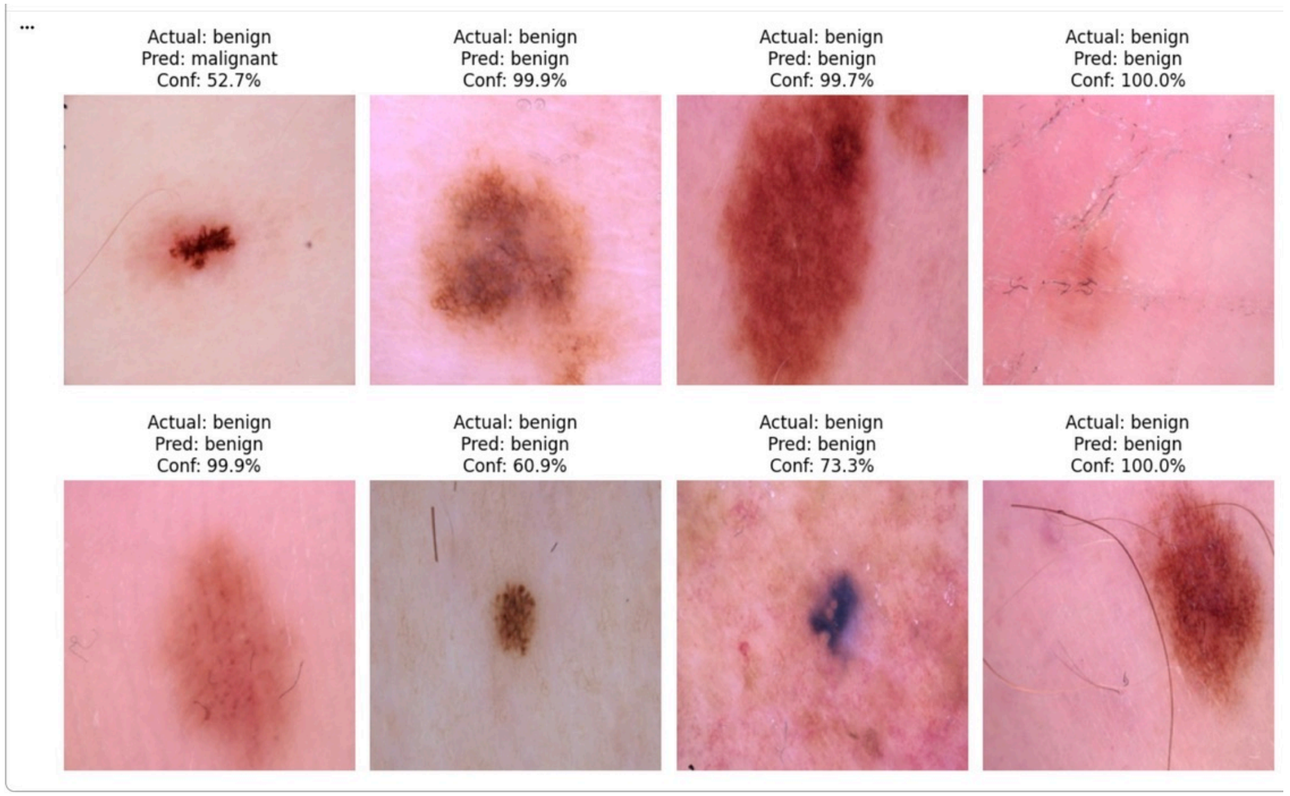
Demo Screenshots

Sample Predictions

Examples include:

- Correct benign predictions
- Correct malignant predictions
- Confidence scores for each prediction

This demonstrates the model working successfully.



Lessons Learned

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This project helped me understand:

- Transfer learning in computer vision
- Model evaluation using confusion matrices
- Importance of data preprocessing
- Hyperparameter tuning for better accuracy
- Real-world applications of image classification

Future Improvements

Future Improvements

Future versions could include:

- Larger medical datasets
- Better models
- Web app deployment for live predictions
- Multi-class lesion classification
- Clinical support tools for dermatologists

This project can be expanded significantly.

Resources

- Google Colab (GPU)
- PyTorch + Torchvision
- Kaggle Dataset
- Python
- Scikit-learn
- Matplotlib
- GitHub
- ChatGPT (project support)
- Total Cost: Free